

Brain Tumor Detection Using YOLOv8 on MRI Images

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Abstract

This paper presents a deep learning approach for detecting brain tumors in MRI images using the YOLOv8 object detection model. The system was trained on a publicly available MRI dataset from Roboflow Universe. The objective is to automatically localize brain tumors with high accuracy and provide a fast computer-aided diagnosis tool.

Keywords

YOLOv8, Brain Tumor, MRI, Object Detection, Artificial Intelligence, Deep Learning.

Introduction

Brain tumors are among the most serious neurological diseases. Early detection can improve diagnosis and treatment planning. Recent advances in computer vision and deep learning have enabled automatic tumor detection from MRI scans. In this project, YOLOv8 was selected because of its speed, accuracy, and real-time detection capability.

Dataset

The dataset used in this work is the Brain Tumor dataset available on Roboflow Universe. It contains annotated MRI images with bounding boxes around tumor regions. The dataset was divided into training, validation, and testing sets before model training.

Methodology

The project was implemented using Python and the Ultralytics YOLOv8 framework in Google Colab. The workflow included dataset preparation, model training, validation, and testing. During inference, the trained model predicts bounding boxes around suspected tumor regions together with confidence scores.

Experimental Results

The trained model achieved the following performance:

Precision: 83.65% Recall:
85.71% mAP@50: 85.26%
mAP@50-95: 68.97%

These results demonstrate that the proposed model is capable of accurately detecting brain tumors in MRI images while maintaining good localization performance.

Discussion

YOLOv8 successfully identified tumor regions with reliable accuracy and fast inference speed. The obtained metrics indicate that the model is suitable for educational and research purposes and can serve as a foundation for future medical AI systems.

Conclusion

This project demonstrates the effectiveness of YOLOv8 for automated brain tumor detection from MRI images. Future work includes training on larger datasets, improving image augmentation, and deploying the model in a desktop or web-based clinical decision support application.

References

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